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Section : BCS-4k

Operating System Lab

Remaining Lab Task 2

Task :2

Code Screenshots:

1.convert.h

```cpp
float celsius_to_fahrenheit(float celsius);

```

2.convert.cpp

```cpp
#include "convert.h"

float celsius_to_fahrenheit(float celsius)
{
    return (celsius*9.0/5.0)+32;
}

```

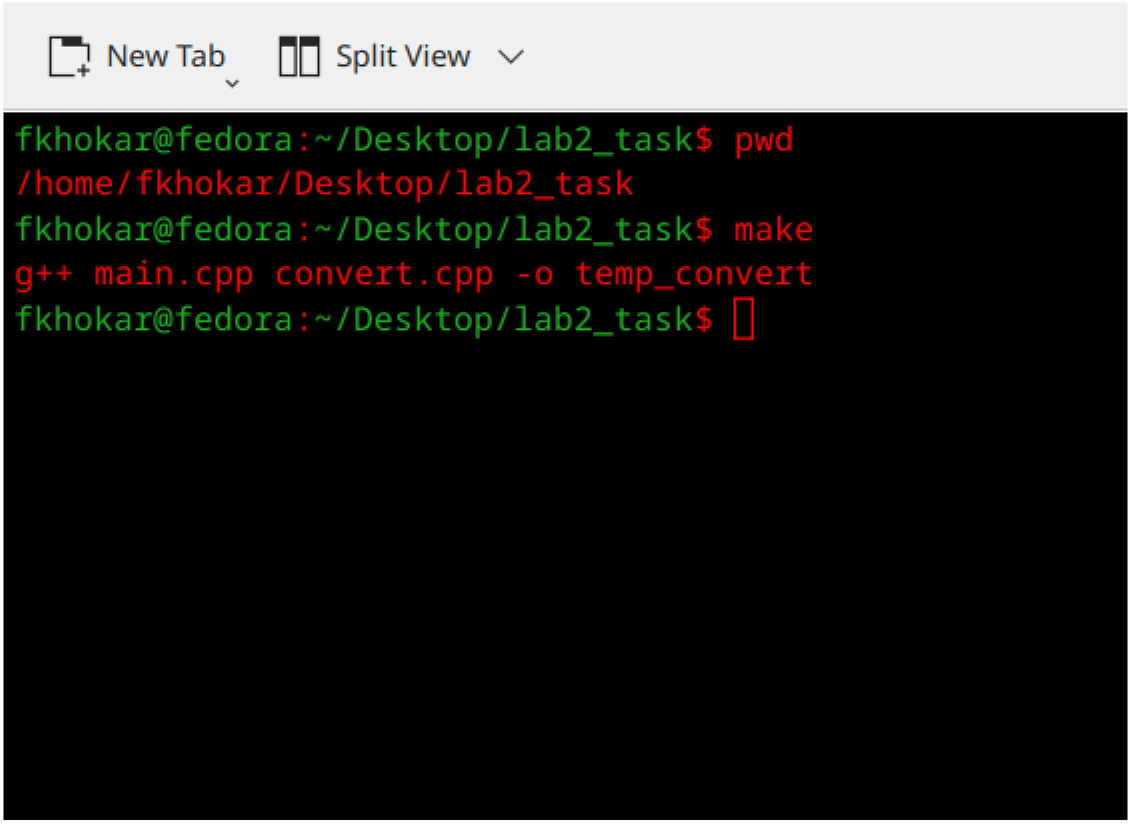
3.main.cpp

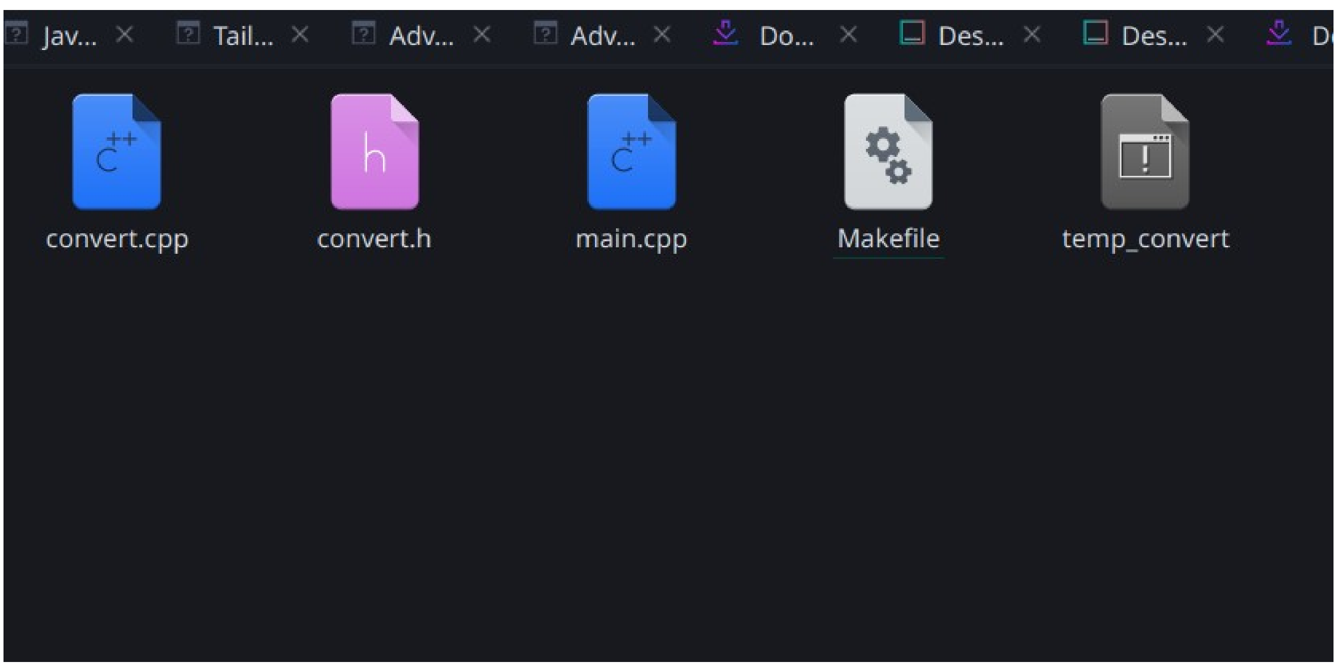
```cpp
#include <iostream>
#include "convert.h"

using namespace std;

int main()
{
    double celsius;

    cout<<"Enter temperature in Celsius: ";
    cin>>celsius;

    cout<<"Temperature in Fahrenheit: " << celsius_to_fahrenheit(celsius)<< endl;

    return 0;
}
```

Make File

```makefile
# Target: this will compile all files into executable temp_convert
temp_convert: main.cpp convert.cpp
	g++ main.cpp convert.cpp -o temp_convert


# Target: this will run the program

run: temp_convert
	./temp_convert


# Target: this will clean the compiled files

clean:
	rm -f temp_convert
```

Make -----Command Screenshot:


New Tab
Split View
fkhokar@fedora:~/Desktop/lab2_task$ pwd
/home/fkhokar/Desktop/lab2_task
fkhokar@fedora:~/Desktop/lab2_task$ make
g++ main.cpp convert.cpp -o temp_convert
fkhokar@fedora:~/Desktop/lab2_task$


Jav...
Tail...
Adv...
Adv...
Do...
Des...
Des...
convert.cpp
convert.h
main.cpp
Makefile
temp_convert

Make run ----Command Screenshot:

File Edit View Bookmarks Plugins Settings Help

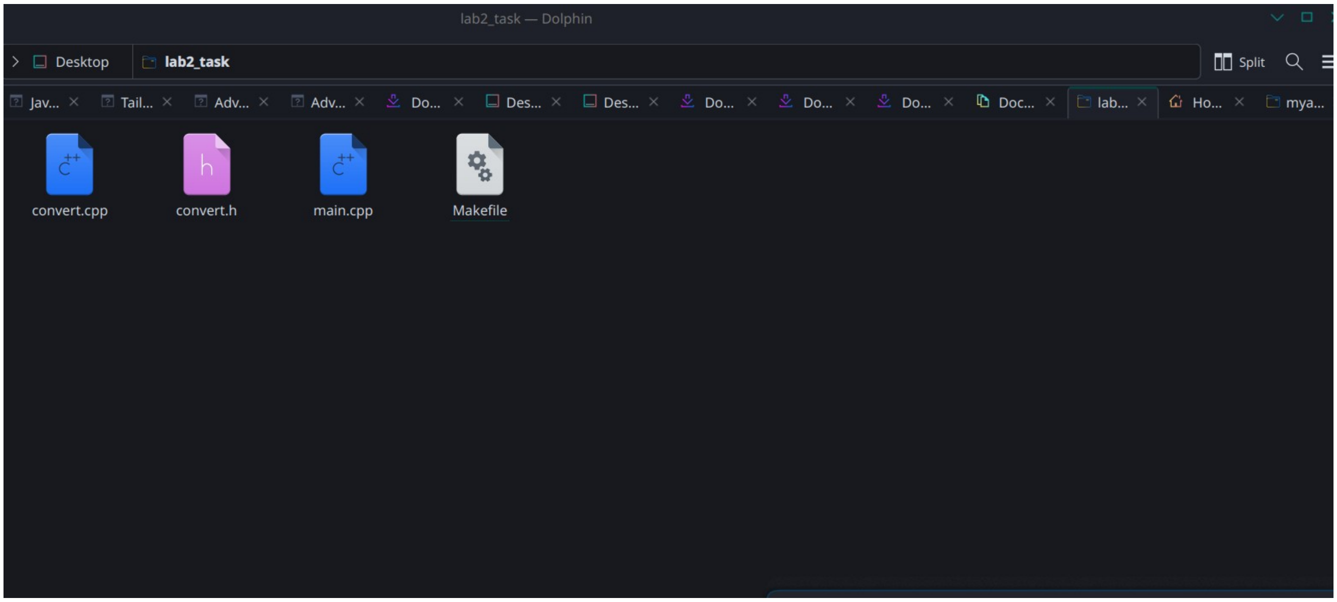
New Tab Split View

```
fkhokar@fedora:~/Desktop/lab2_task$ pwd
/home/fkhokar/Desktop/lab2_task
fkhokar@fedora:~/Desktop/lab2_task$ make
g++ main.cpp convert.cpp -o temp_convert
fkhokar@fedora:~/Desktop/lab2_task$ make run
./temp_convert
Enter temperature in Celsius: 32
Temperature in Fahrenheit: 89.6
fkhokar@fedora:~/Desktop/lab2_task$
```

Make clean---- command screenshot:

```
fkhokar@fedora:~/Desktop/lab2_task$ make clean
rm -f temp_convert
fkhokar@fedora:~/Desktop/lab2_task$
```

Excutable file deleted by make clean ---command:



Task : 3

Codes screenshots

1.checkinvalid.h

2.checkinvalid.cpp

Makefile × checkinvalid.h × checkinvalid.cpp × main.cpp ×

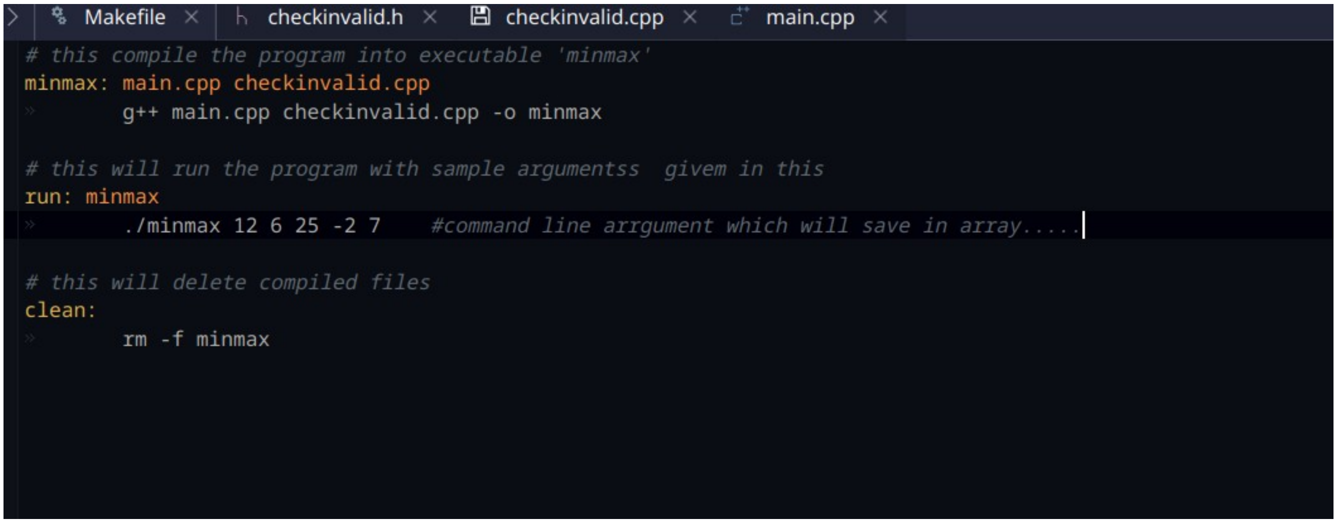
```cpp
#include "checkinvalid.h"


bool is_valid_number(char* string) // This function willl checks whether the input string  is a valid integer numbers
{
    int i = 0;

    // this will allow negative number from arguments.........
    if (string[0] == '-')
        i = 1;


    if (string[i] == '\0')   // string should not be empty after sign like 123- it should be 123-4
        return false;

    while (string[i] != '\0')
    {
        if (string[i] < '0' || string[i] > '9')
            return false;
        i++;
    }
    return true;
}


int string_to_integer(char* string)    //This function will converts a numeric string givien through command line in make file into an integer
{
    int i = 0, num = 0, sign = 1;

    if (string[0] == '-')
    {
        sign = -1;
        i = 1;
    }

    while (string[i] != '\0')
    {
        num = num * 10 + (string[i] - '0');
        i++;
    }

    return num * sign;
}
```

3.main.cpp

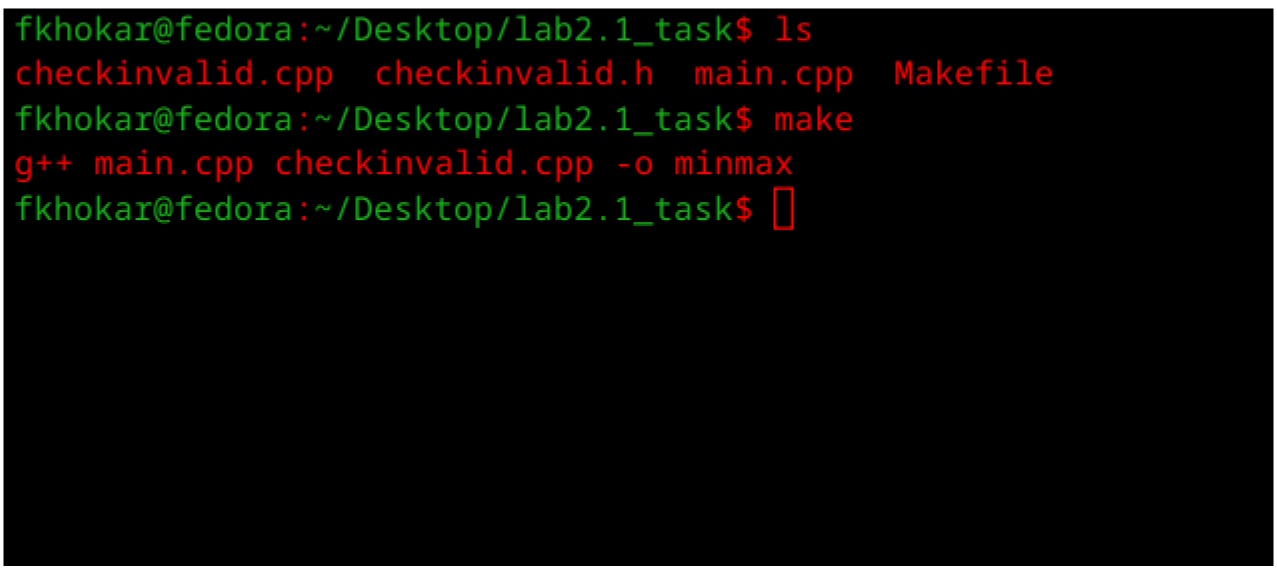
Makefile × checkinvalid.h × checkinvalid.cpp × main.cpp ×

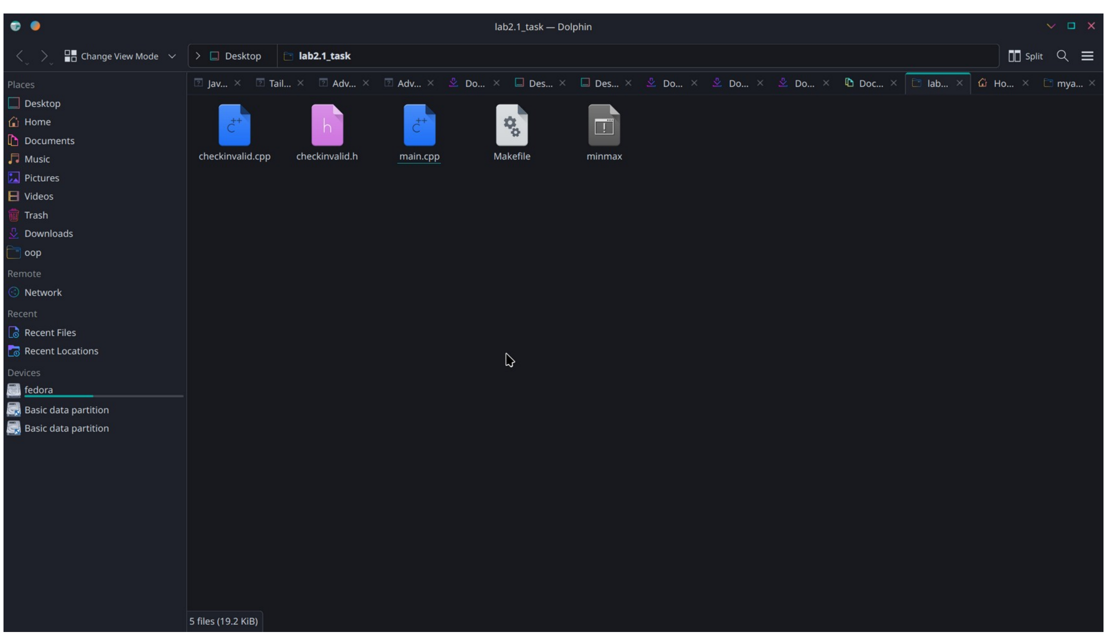
```cpp
#include <iostream>
#include "checkinvalid.h"

using namespace std;

int main(int argc, char* argv[])       //argc is for number of inutes and arv is for what value is......
{
    if (argc < 2)
    {
        cout << "No command line arguments are provided." << endl;
        return 1;
    }

    int arr[argc - 1];

    for (int i = 1; i < argc; i++)
    {
        if (!is_valid_number(argv[i]))
        {
            cout << "Error: Invalid input: " << argv[i] << endl;
            return 1;
        }
        arr[i - 1] = string_to_integer(argv[i]);
    }

    int maximum = arr[0];
    int minimum = arr[0];

    for (int i = 1; i < argc - 1; i++)
    {
        if (arr[i] > maximum)
            maximum = arr[i];
        if (arr[i] < minimum)
            minimum = arr[i];
    }

    cout << "Maximum number is : " << maximum << endl;
    cout << "Minimum number is : " << minimum << endl;

    return 0;
```

Make file screenshot:



Make ----command screenshot:





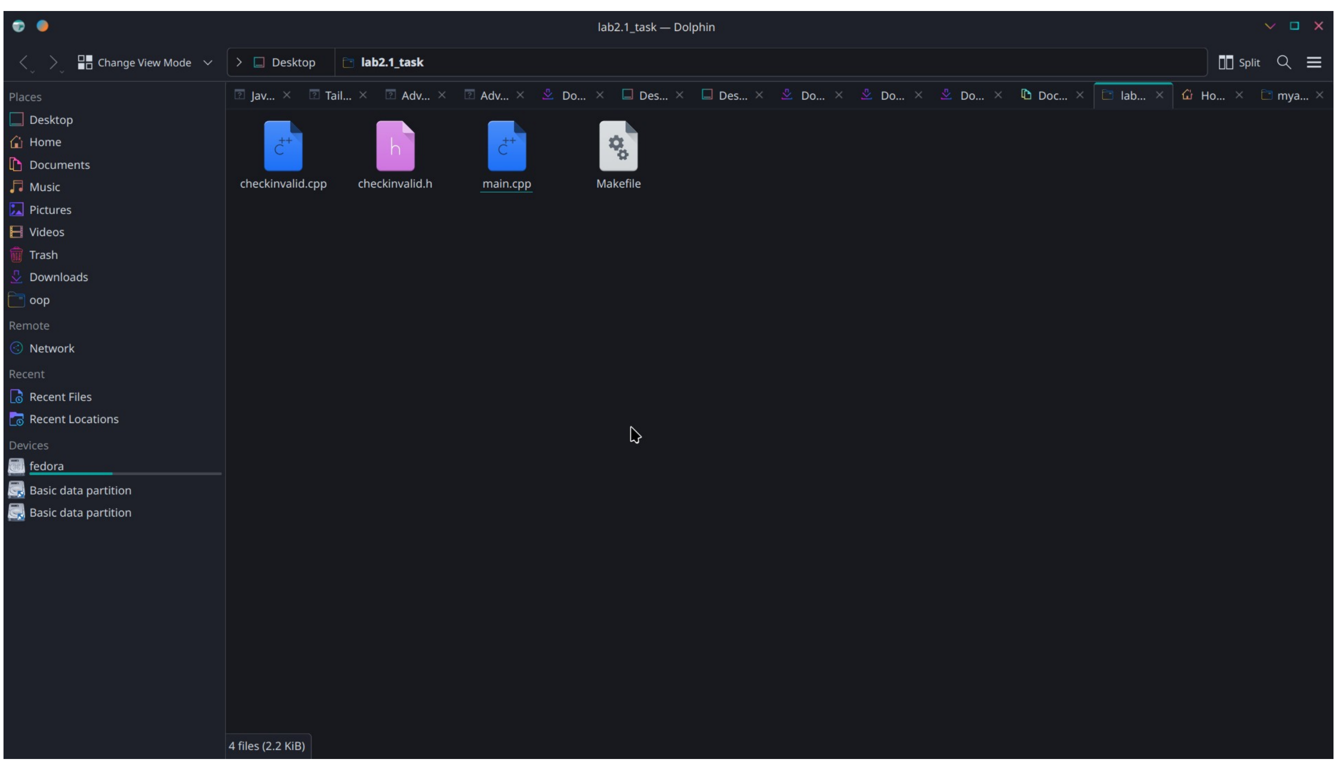
Make run ----Command screenshot

```
fkhokar@fedora:~/Desktop/lab2.1_task$ make run
./minmax 12 6 25 -2 7    #command line arrgument which will save in array.....
Maximum number is : 25
Minimum number is : -2
fkhokar@fedora:~/Desktop/lab2.1_task$ 
```

Make clean -----command sceenshot

```
fkhokar@fedora:~/Desktop/lab2.1_task$ make clean
rm -f minmax
fkhokar@fedora:~/Desktop/lab2.1_task$ 
```

Excutable file is deleted by make clean -----command



Show error when no arguments are passed below is screenshots

```
g++ main.cpp checkinvalid.cpp -o minmax
fkhokar@fedora:~/Desktop/lab2.1_task$ make run
./minmax
No command line arguments are provided.
make: *** [Makefile:7: run] Error 1
fkhokar@fedora:~/Desktop/lab2.1_task$
```

Makefile × checkinvalid.h × checkinvalid.cpp × main.cpp ×

```
# this compile the program into executable 'minmax'
minmax: main.cpp checkinvalid.cpp
	g++ main.cpp checkinvalid.cpp -o minmax

# this will run the program with sample argumentss  givem in this
run: minmax
	./minmax

# this will delete compiled files
clean:
	rm -f minmax

```

Sow error when non numeric value is provided:

```
fkhokar@fedora:~/Desktop/lab2.1_task$ make run
./minmax a b c    #command line arrgument which will save in array.....
Error: Invalid input: a
make: *** [Makefile:7: run] Error 1
fkhokar@fedora:~/Desktop/lab2.1_task$
```

Makefile × checkinvalid.h × checkinvalid.cpp × main.cpp ×

```
# this compile the program into executable 'minmax'
minmax: main.cpp checkinvalid.cpp
	g++ main.cpp checkinvalid.cpp -o minmax

# this will run the program with sample argumentss  givem in this
run: minmax
	./minmax a b c    #command line arrgument which will save in array.....

# this will delete compiled files
clean:
	rm -f minmax

```